

Lesson 14 — The Machine Answers Back

Matches LEGO Teacher Portal lesson: "Treat Time".

Objective (I can...): I can make the hardware answer an event right away — the sensor or press makes the motor respond. **CS concept:** Hardware responding to events · **Builds on:** Lesson 13 (creating events).

AI Connection (2 min, read aloud): An always-on helper waits for an event and answers the moment it happens. *Ask the group:* "Who does an always-listening speaker help? Could it bother anyone at home?"

Materials: LEGO CS & AI kit + device per pair, color cards/trigger objects. **Model reuse:** Events model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Answer back" — coach gives a signal (raise a hand), kids instantly respond (stand up). The faster the better. Event in, response out. - **Concept + demo (8):** Demo an event that makes the motor respond right away — press the button, the motor spins; show the color sensor event making the motor move. Hardware is answering the event. - **Build/Code (25):** Pairs code events so the motor (and a light or sound) responds the instant a trigger happens — press → motor spins; color seen → motor reverses. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a second event so the machine has two different hardware responses. - **Share + tidy-up (10):** Pairs trigger each event and show the motor answering. Model stays built.

Pair roles: Driver codes, Navigator triggers the event and watches the motor respond. Swap at the wiggle break. (*Trio rotates code/trigger/watch.*)

Differentiation: *Support* — one event → one motor response. *Extension / fast-finisher* — two events with two different motor responses. *Reteach* — press and watch nothing until the event block is added, then add it together.

Exit check: Pair's hardware (motor) responds correctly the moment the event happens. **At-home (optional):** Find something that answers right away when you do something (a light switch, a clap-on lamp); tell a grown-up the event and the response. ~5 min.
